

HR Analytics Dashboard

Final Analysis Report By Sourav Roychoudhury

Tools: Power BI | Power Query | DAX | Microsoft Excel

Executive Summary

The HR Analytics Dashboard is an end-to-end Business Intelligence project developed using Power BI, Power Query, DAX, and Microsoft Excel to analyze employee attendance patterns and workforce availability. The project simulates a real-world HR analytics scenario where business requirements are gathered from HR stakeholders and translated into an interactive dashboard for strategic decision-making.

The dashboard provides comprehensive insights into employee Presence Percentage, Work From Home (WFH) trends, Sick Leave (SL) patterns, and attendance behavior across different time periods and weekdays. The project demonstrates how Business Intelligence can transform raw attendance records into meaningful insights that support data-driven HR decisions.

1. Project Objective

The primary objective is to build an interactive HR Analytics Dashboard that enables organizations to monitor employee attendance and workforce availability. The project addresses key business questions such as:

- What is the overall employee presence percentage?
- How frequently are employees working from home?
- What is the sick leave percentage across the organization?
- Are there any attendance patterns across weekdays?
- Which employees have higher absenteeism?
- How can HR improve workforce planning using attendance insights?

2. Dataset Overview

The project uses a masked employee attendance dataset stored in Microsoft Excel. Each attendance record represents the daily working status of an employee. Key fields include:

Field	Description
Employee Name	Full name of the employee
Employee Code	Unique employee identifier

Attendance Date	Date of the attendance record
Presence Status	Whether the employee was present
Work From Home Status	Whether the employee worked remotely
Sick Leave Status	Whether the employee was on sick leave
Attendance Codes	Coded values representing attendance type
Month / Weekday	Time dimension attributes

3. Business Requirement Gathering

The project begins with understanding business requirements from an HR stakeholder. The primary reporting requirements include:

- Monitoring employee attendance percentage
- Measuring Work From Home utilization
- Tracking Sick Leave trends
- Comparing attendance across weekdays
- Creating employee-level attendance summaries
- Supporting workforce planning and HR policy decisions

This requirement gathering phase ensures that the dashboard aligns with real business needs rather than purely technical metrics.

4. Data Cleaning & Transformation

Power Query was used extensively to prepare the attendance dataset for analysis.

Data Cleaning

- Removed unnecessary columns
- Corrected inconsistent attendance values
- Standardized attendance codes
- Converted date fields into proper date format

Unpivot Transformation

Attendance columns were unpivoted into a normalized table structure, allowing efficient analysis across dates.

Date Dimension Creation

Additional calendar attributes were created including Month, Weekday, Week Number, and Day to improve reporting flexibility.

5. Data Modeling

A structured data model was developed to support efficient reporting. The model connects three tables:

Table	Description
Attendance Table	Daily attendance records for all employees
Calendar Table	Date dimension with month, weekday, and week attributes
Employee Table	Employee profiles and reference information

Relationships between these tables allow dynamic filtering and accurate KPI calculations across the dashboard.

6. DAX Measures

Several calculated measures were created using DAX to evaluate workforce performance. Key KPIs include:

KPI	Description
Presence %	Percentage of working days employees were physically present
Work From Home %	Percentage of days employees worked remotely
Sick Leave %	Percentage of attendance records marked as sick leave
Attendance Counts	Total attendance records for reporting purposes

These dynamic calculations automatically update based on dashboard filters and selections.

7. Dashboard Overview

The Power BI dashboard provides an interactive overview of workforce attendance through the following components:

- KPI Cards showing Presence %, WFH %, and Sick Leave %
- Area Charts for trend analysis over time
- Employee-Level Attendance Tables
- Weekday Breakdown visuals
- Interactive Date and Employee Filters

- Cross-filtering and drill-down functionality

Users can drill down into attendance trends across different employees and time periods without modifying the underlying dataset.

8. Key Analysis Areas

Employee Presence Analysis

Presence Percentage is the primary KPI displayed on the dashboard. It calculates the proportion of days employees were physically present at work. Higher presence percentages generally indicate greater workforce availability. This metric enables HR to evaluate workforce availability, identify absenteeism trends, monitor employee engagement, and improve office resource planning.

Work From Home (WFH) Analysis

The dashboard tracks WFH utilization across the organization, revealing percentage of employees working remotely, monthly WFH trends, and weekday-based WFH patterns. Increasing WFH percentages may indicate adoption of hybrid work policies, greater workplace flexibility, and changing employee work preferences.

Sick Leave Analysis

Sick Leave Percentage is calculated using attendance records marked as sick leave. The dashboard identifies total sick leave utilization, monthly trends, and weekday distribution. This helps organizations detect seasonal illness patterns, forecast staffing shortages, improve workforce planning, and evaluate employee well-being.

Weekday Analysis

Attendance metrics are analyzed across each day of the week — Monday through Friday. The analysis compares Presence %, WFH %, and Sick Leave % by weekday. Example patterns observed include:

- Slightly lower office attendance on Mondays and Fridays
- Higher Work From Home utilization near weekends
- Sick leave clustering on specific weekdays

These insights support workforce scheduling decisions and office occupancy planning.

Employee-Level Performance

The dashboard includes a detailed employee attendance table where each employee can be evaluated based on Presence %, WFH %, Sick Leave %, and full Attendance History. HR managers can identify high performers, frequent absentees, employees with excessive sick leave, and employees heavily utilizing WFH.

Key Business Insights

- The dashboard provides a centralized view of employee attendance, WFH utilization, and sick leave trends
- Presence percentage serves as a key indicator of workforce availability and overall organizational productivity
- WFH analysis highlights evolving hybrid work patterns and helps optimize workplace resource allocation
- Sick leave monitoring enables HR teams to identify seasonal trends, anticipate staffing shortages, and support employee wellness
- Weekday analysis reveals recurring attendance behaviors that inform scheduling and office occupancy planning
- Employee-level insights allow HR managers to identify irregularities, recognize consistent performers, and address absenteeism proactively
- Interactive filtering and dynamic KPIs enable stakeholders to perform detailed analysis without manual data processing

Business Recommendations

1. **Optimize Hybrid Work Policies:** Leverage WFH trends to establish balanced hybrid work schedules that maintain productivity while supporting employee flexibility.
2. **Improve Attendance Monitoring:** Regularly review employee presence percentages to identify absenteeism early and implement appropriate interventions.
3. **Monitor Sick Leave Patterns:** Analyze sick leave trends to anticipate seasonal workforce shortages and develop proactive staffing strategies.
4. **Strengthen Workforce Planning:** Use attendance and presence data to forecast staffing requirements, improve office space utilization, and allocate resources efficiently.
5. **Encourage Employee Engagement:** Recognize employees with strong attendance records and investigate recurring absenteeism to improve workforce morale and performance.
6. **Expand HR Analytics:** Enhance the dashboard by integrating metrics such as employee turnover, overtime, performance evaluations, leave balances, and satisfaction scores for a more comprehensive workforce analytics solution.

Skills Demonstrated

Category	Skills
BI & Visualization	Microsoft Power BI, Dashboard Development, Interactive Data Visualization
Data Transformation	Power Query (M Language), Data Cleaning, Unpivot Transformation
Calculations	DAX Measure Development, KPI Design, Dynamic Calculations

Analytics	HR Analytics, Workforce Analytics, Attendance Analysis, Performance Monitoring
Workflow	Data Modeling, Business Intelligence, Data Storytelling, End-to-End Analytics

Conclusion

The HR Analytics Dashboard demonstrates how Business Intelligence can convert raw employee attendance data into actionable workforce insights that support strategic HR decision-making. By integrating Power Query for data transformation, DAX for dynamic KPI calculations, and Power BI for interactive visualization, the project delivers a comprehensive solution for monitoring attendance, hybrid work adoption, and employee availability.

The dashboard enables HR teams to optimize workforce planning, evaluate attendance policies, improve operational efficiency, and make informed decisions based on real-time analytics. As an end-to-end HR analytics solution, the project showcases strong capabilities in data preparation, modeling, visualization, and business storytelling, making it a valuable portfolio project for data analytics and business intelligence roles.